

Driven Business Information Technology undergraduate at Strathmore University on track for First Class Honours and a three-time Dean's List recipient. Strong interest in software engineering, machine learning, and product development. Experience building machine learning systems, web platforms, and secure messaging applications using modern engineering tools and frameworks.

EDUCATION

Strathmore University

Nairobi, Kenya

Bachelor of Business Information Technology – Expected First Class Honours

Expected July 2026

- **Relevant Coursework:** Software Engineering, Object-Oriented Programming, Data Structures & Algorithms, Procedural Programming, Computer Organisation & Architecture, Cryptography & Network Security
- **Application Development:** Web Application Development, Mobile Application Development, Desktop Application Development
- **Data & Intelligence:** Database Management, Business Intelligence, Artificial Intelligence & Machine Learning
- **Business:** Cost & Financial Accounting, Business Finance, Economics, Management, Marketing, Business Process Management, Business Systems Analysis, IT Project Management

WORK EXPERIENCE

Old Mutual Kenya

Nairobi, Kenya

IT & Network Infrastructure Intern

Jan 2025 – Mar 2025

- Supported the ICT department in optimizing internal systems using Python and SQL for automation and data processing.
- Configured analytics tools to aggregate network logs and operational data, improving incident response time by roughly 30%.
- Containerized internal scripts with Docker and maintained Git-based version control for infrastructure utilities.
- Participated in security and compliance audits aligned with ISO/IEC 27001 standards.
- Helped integrate on-premise data sources with cloud dashboards for hybrid analytics reporting.

Strathmore Joint Information Systems Project

Nairobi, Kenya

Head of System Operations (Back-End Development)

Apr 2024 – Jul 2024

- Led back-end development for a QR-based digital attendance tracking system deployed across departments.
- Built REST APIs using Node.js and PostgreSQL to record and visualize attendance data in real time.
- Implemented QR token verification modules using cryptographic signatures to prevent spoofing.
- Designed a Python ETL pipeline integrating attendance logs into Power BI dashboards.
- Containerized the system with Docker and implemented CI/CD pipelines using GitHub Actions.

McMillan Memorial Library

Nairobi, Kenya

Head Research Assistant

Jan 2024 – Mar 2024

- Led a digitization and NLP metadata classification project covering more than 15,000 archival records.
- Built a Python and MySQL ETL pipeline for ingestion and validation of archival datasets.
- Developed a document classification prototype using scikit-learn for automated tagging.
- Applied NLP techniques for metadata enrichment, improving classification precision by 64%.
- Collaborated with postgraduate researchers on data labeling and model evaluation.

PROJECTS

- **ML-Powered Exam Anomaly Detection System:** Built a Python-based ML pipeline using scikit-learn and TensorFlow to detect anomalies in exam results. Implemented preprocessing and normalization achieving 92% model accuracy. Integrated analytics dashboards with Dash and Power BI for administrative insights.
- **Personal Online Portfolio:** Developed and deployed nduatileon.site using Next.js, React, and TailwindCSS. Implemented dynamic project and music pages with version control through GitHub and CI/CD deployment via Vercel.
- **Encrypted Messaging Web Application:** Designed a Flask-based secure messaging platform with AES encryption and OTP authentication using Infobip SMS integration. Implemented RESTful encryption and decryption APIs with a simple web interface.

HONORS & AWARDS

- **Dean's List Award (2023–2025):** Strathmore University academic distinction.
- **Digital Explorers II Hackathon – Top 4 Projects:** Recognized among leading projects during the November 2025 hackathon.
- **IBM Certificate – Artificial Intelligence and Machine Learning Fundamentals:** Certification covering foundations of machine learning and AI systems.
- **Safaricom SCES Certificate – Software Architecture Design and Automation:** Professional training in scalable software architecture.

SKILLS

Programming Languages: Python, JavaScript, C++, Java, Kotlin, C#
Frameworks & Libraries: Next.js, React, Laravel, Flask, Unity, TailwindCSS
AI / Machine Learning: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, NLTK, spaCy
Cloud & DevOps: Docker, IBM Cloud, Git, GitHub Actions, Vercel
Core Competencies: Data Structures & Algorithms, REST APIs, Feature Engineering, Natural Language Processing, Model Evaluation, Agile Development, Version Control

COMMUNITY SERVICE

- Strathmore Environmental Sustainability Club**
Mazingira Day Volunteer

Nairobi, Kenya
Oct 2025

 - Participated in university tree-planting initiatives supporting campus reforestation.
 - Logged planted species and locations to support biodiversity tracking efforts.
 - Attended environmental awareness sessions on sustainability and renewable energy.
- Limuru Maraba Tea Farm**
Community Outreach Volunteer

Limuru, Kenya
Jun 2024 – Jul 2024

 - Helped coordinate food and clothing donations for tea farm worker families.
 - Supported renovation of a kindergarten classroom to improve early learning facilities.

LANGUAGES

- **German:** Intermediate
- **French:** Intermediate
- **Greek:** Intermediate
- **Hebrew:** Beginner